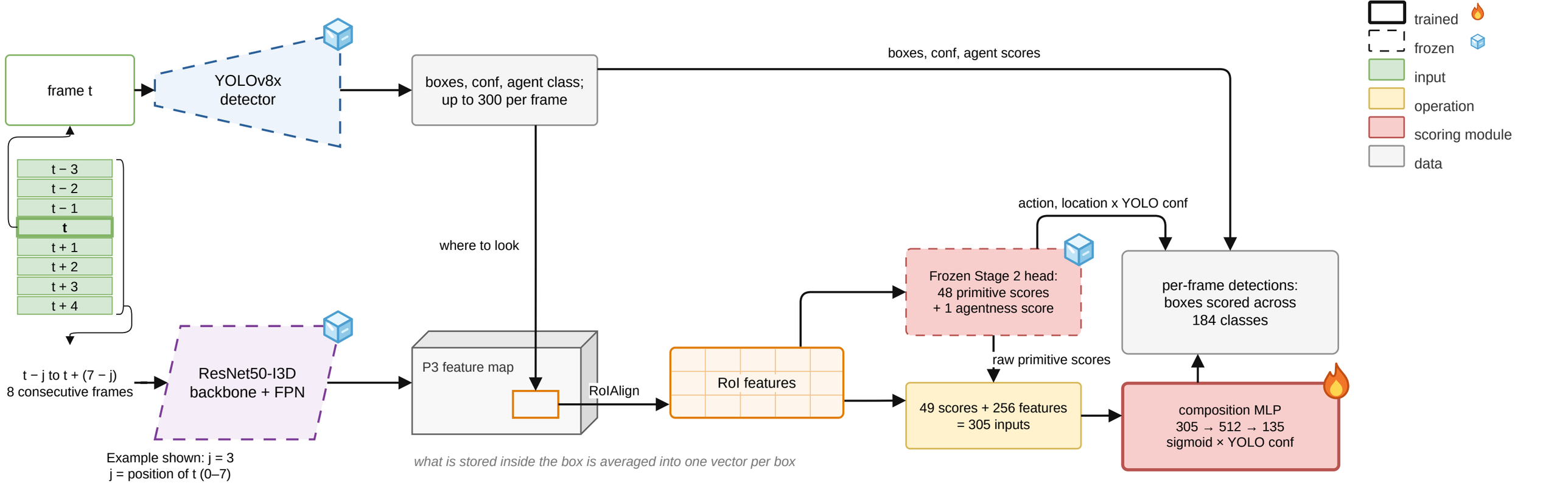

frame t
YOLOv8x detector
boxes, conf, agent class; up to 300 per frame
boxes, conf, agent scores
t − 3
t − 2
t − 1
t
t + 1
t + 2
t + 3
t + 4
where to look
action, location x YOLO conf
Frozen Stage 2 head: 48 primitive scores + 1 agentness score
per-frame detections: boxes scored across 184 classes
t − j to t + (7 − j)
8 consecutive frames
ResNet50-I3D backbone + FPN
P3 feature map
RoIAlign
RoI features
raw primitive scores
49 scores + 256 features = 305 inputs
composition MLP 305 → 512 → 135 sigmoid × YOLO conf
Example shown: j = 3
j = position of t (0–7)
what is stored inside the box is averaged into one vector per box
trained
frozen
input
operation
scoring module
data